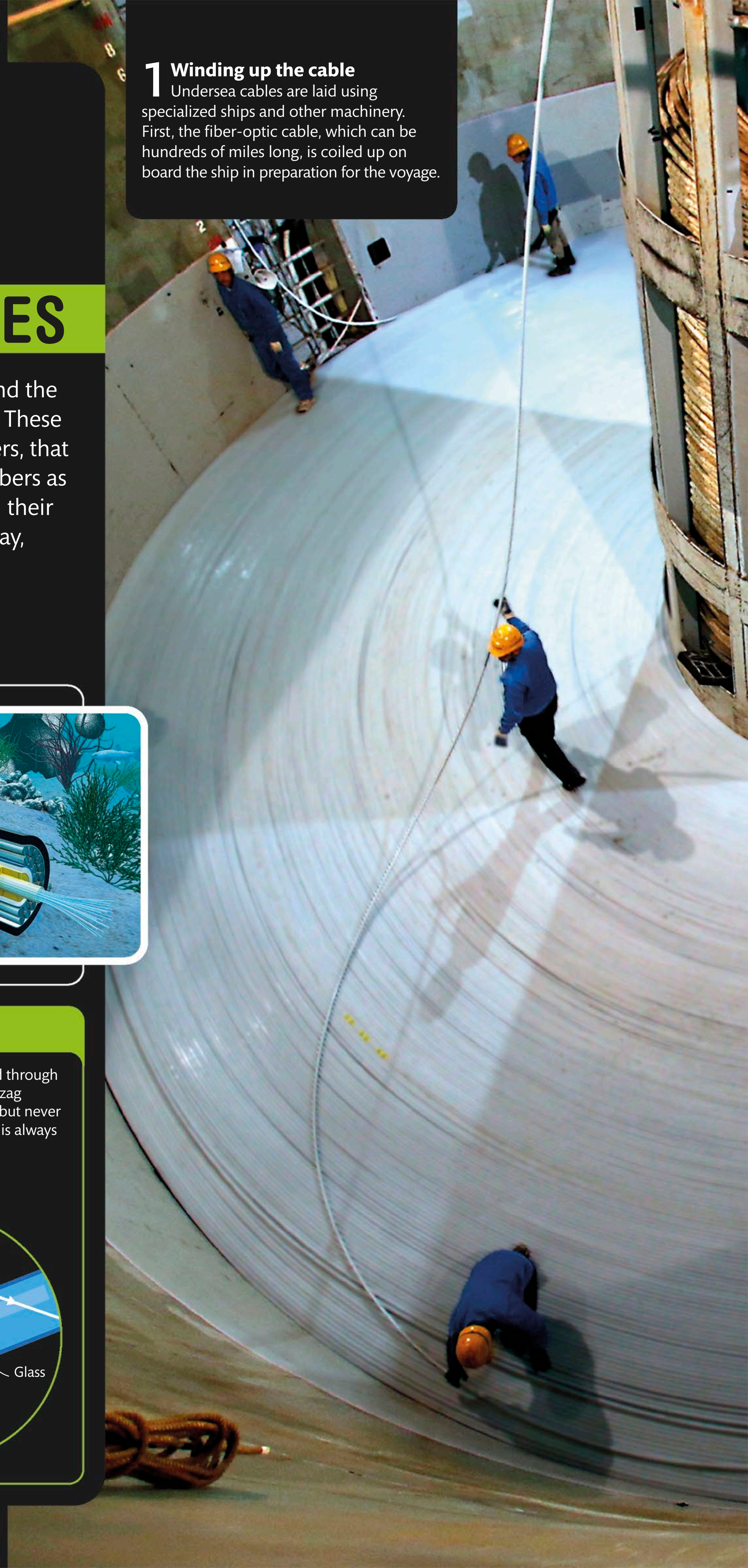
DELIVERING DATA

FIBER-OPTIC CABLES

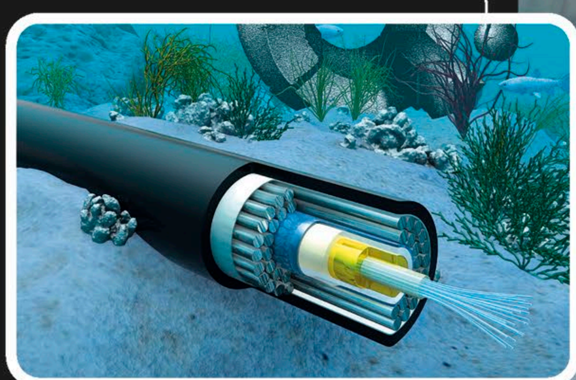
Every day, vast amounts of data are sent around the globe safely and quickly using fiber-optic cables. These cables contain strands of glass, called optical fibers, that are as thin as a hair. Data speeds through these fibers as pulses of light, reflecting back and forth between their walls as it races from continent to continent. Today, the internet, television, and telephone systems all rely on fiber-optic cable networks.



1 Winding up the cable Undersea cables are laid using specialized ships and other machinery. First, the fiber-optic cable, which can be hundreds of miles long, is coiled up on board the ship in preparation for the voyage.

INSIDE A CABLE

A fiber-optic cable contains optical fibers, individually wrapped in a layer called cladding. The fibers are then bundled together and sealed in a tube to protect the cables when they are laid out in exposed environments, such as the seabed.



LIGHT AND REFLECTION

In optical fibers, information is carried in light signals. Light is transferred through the fibers by a process called total internal reflection, as it travels in a zigzag pattern. The light wave bounces from one side of the cable to the other but never hits the edge of the glass at an angle steep enough to pass through, so it is always completely reflected—all the way through the length of the cable.